Do Agents Look for Truth or Comfort? Measuring the Truth Discount in LLM Agent Societies

Anonymous submission

Abstract

1 Introduction

2 Related Work

3 Methodology

4 Experiments

5 Results

6 Discussion

Ethical Statement

As LLM agent networks are increasingly trusted with automated workflows and shared deliberation, they risk amplifying ungrounded claims simply to keep conversations agreeable or defer to perceived authority. Understanding how and why artificial agents sacrifice factual truth for social comfort is a necessary step toward building effective verification checks and auditing systems.

To explore this safely without generating harmful blueprints for real-world manipulation, we intentionally restricted our experiments to low-stakes community concert rumors with invented artist names. We avoided politically sensitive, medical, or financial topics to prevent misuse. Furthermore, our experiments consist solely of simulated agent-to-agent interactions; no human subjects were exposed to false claims or deceptive interactions, and no private personal records were collected or processed.

References

Borah, A.; Mihalcea, R.; and Pérez-Rosas, V. 2026. Persuasion at Play: Understanding Misinformation Dynamics in Demographic-Aware Human-LLM Interactions. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 5027–5053.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

Chuang, Y.-S.; Goyal, A.; Harlalka, N.; Suresh, S.; Hawkins, R.; Yang, S.; Shah, D.; Hu, J.; and Rogers, T. 2024. Simulating Opinion Dynamics with Networks of LLM-based Agents. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 3326–3346.

Kumar, A. R.; and Chopra, P. 2026. Authority Bias in Language Models: Source Deference and User Agreement Are Not Interchangeable. arXiv:2609.37616.

Liu, Y.; Song, Z.; Zhang, J.; Zhang, X.; Chen, X.; and Yan, R. 2025. The Stepwise Deception: Simulating the Evolution from True News to Fake News with LLM Agents. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 26176–26192.

Mahmud, S.; Bagdasarian, E.; and Zilberstein, S. 2026. Verification Required: The Impact of Information Credibility on AI Persuasion. arXiv:2602.00970.

Münker, S.; Schwager, N.; and Rettinger, A. 2026. Don't Trust Generative Agents to Mimic Communication on Social Networks Unless You Benchmarked their Empirical Realism. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 1141–1151.

Park, J. S.; O'Brien, J. C.; Cai, C. J.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2023. Generative Agents: Interactive Simulacra of Human Behavior. arXiv:2304.03442.

Schuster, J.; Gautam, V.; and Markert, K. 2026. Whose Facts Win? LLM Source Preferences under Knowledge Conflicts. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 29430–29459.

A Experimental Protocol and Simulation Details

B Declaration of Generative AI and AI-Assisted Technologies

We used foundation model assistants (Google Gemini and Anthropic Claude) to help with manuscript preparation and development support. Their assistance was limited to drafting boilerplate code for running trials and plotting figures, organizing bibliography citations, and polishing text readability and LaTeX formatting.

All primary scientific contributions—including the research questions, the counterbalanced twin-world design, the mathematical definition of the Truth Discount, the experimental runs, and the empirical findings—were designed, analyzed, and evaluated by the human authors. The authors are fully responsible for the contents, claims, and conclusions of this paper.

Reproducibility Checklist

Instructions for Authors:

This document outlines key aspects for assessing reproducibility. Please provide your input by editing this `.tex` file directly.

For each question (that applies), replace the "Type your response here" text with your answer.

Example: If a question appears as

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
Type your response here
```

you would change it to:

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
yes
```

Please make sure to:

- Replace ONLY the "Type your response here" text and nothing else.
- Use one of the options listed for that question (e.g., **yes**, **no**, **partial**, or **NA**).
- **Not** modify any other part of the `\question` command or any other lines in this document.

You can `\input` this .tex file right before `\end{document}` of your main file or compile it as a stand-alone document. Check the instructions on your conference's website to see if you will be asked to provide this checklist with your paper or separately.

---

1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) Type your response here

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) Type your response here

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) Type your response here

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no) Type your response here

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) Type your response here

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) Type your response here

2.4. Proofs of all novel claims are included (yes/partial/no) Type your response here

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) Type your response here

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) Type your response here

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) Type your response here

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) Type your response here

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no) Type your response here

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) Type your response here

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) Type your response here

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) Type your response here

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) Type your response here

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) Type your response here

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) Type your response here

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no) Type your response here

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) Type your response here

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) Type your response here

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) Type your response here

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) Type your response here

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) Type your response here

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) Type your response here

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) Type your response here

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) Type your response here

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) Type your response here

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) Type your response here

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) Type your response here

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) Type your response here